THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The three points $O(0,0)$, $A(3,4)$ and $B(-3,-4)$: 1

(a) form a right triangle **(b)** form an equilateral triangle
(c) form an isosceles triangle **(d)** are collinear

2. The midpoint of the segment joining $A(0,-8)$ and $B(6,0)$ is: 1

(a) $(3,-4)$ **(b)** $(3,4)$ **(c)** $(6,-8)$ **(d)** $(4,-3)$

3. $M(3,2)$ is the midpoint of AB and $A(1,-4)$. The coordinates of B are: 1

(a) $(5,8)$ **(b)** $(4,-2)$ **(c)** $(5,-8)$ **(d)** $(2,6)$

4. The point $(7,15)$ ___ the line $y = 2x + 1$. 1

(a) is the y-intercept of the line **(b)** does not lie on the line
(c) is the origin **(d)** lies on the line

5. For the line $y = x + 3$, the slope and the y-intercept are respectively: 1

(a) 1 and -3 **(b)** 1 and 3
(c) -1 and 3 **(d)** 3 and 1

6. $0.999\ldots$ ($0.\overline{9}$) is equal to: 1

(a) 0.9 **(b)** slightly less than 1
(c) 1 **(d)** 0.99

7. The value of $(-12) \times 5$ is: 1

(a) -60 **(b)** 60 **(c)** -58 **(d)** 59

8. A rational number in lowest terms has denominator $2^2 \times 5^4$. Its decimal expansion terminates after how many places? 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

9. If $a = 10$ and $b = 2$, then $(a+b)^2 - (a^2 + b^2)$ equals: 1

(a) 20 **(b)** 0 **(c)** 40 **(d)** 144

10. The value of 99^2 using a suitable identity is: 1

(a) 9801 **(b)** 9821 **(c)** 9901 **(d)** 9791

11. In a circle of radius 10 cm, a chord is 8 cm from the centre. The length of the chord is: 1

(a) 14 cm **(b)** 12 cm **(c)** 10 cm **(d)** 6 cm

12. The angle subtended by a diameter at any point on the circle is: 1

(a) 60° **(b)** 45° **(c)** 180° **(d)** 90°

13. The diameter of a circle is 26 cm and a chord is 24 cm long. The distance of the chord from the centre is: 1

(a) 5 cm **(b)** 1 cm **(c)** 25 cm **(d)** 13 cm

14. The area of a triangle with base 24 cm and height 5 cm is: 1

(a) 60 cm^2 **(b)** 84 cm^2 **(c)** 29 cm^2 **(d)** 120 cm^2

15. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 14 cm is: 1

(a) 308 cm^2 **(b)** 616 cm^2 **(c)** 88 cm^2 **(d)** 1232 cm^2

16. A median of a triangle divides it into two triangles of: 1

(a) equal area **(b)** one double the other
(c) equal angles **(d)** equal perimeter

17. A fair die is rolled once. The probability of getting an even number is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

18. A letter is picked at random from the word ‘MATHEMATICS’. The probability of picking the letter M is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{2}{11}$ **(d)** $\frac{9}{11}$

19. The number of black triangles at Stage n of the Sierpiński triangle is: 1

(a) $3n$ **(b)** 4^n **(c)** 2^n **(d)** 3^n

20. The sum of the first 20 natural numbers is: 1

(a) 210 **(b)** 209 **(c)** 420 **(d)** 211

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. Write the coordinates of the point on the y-axis that is 6 units above the origin and of the point on the x-axis that is 4 units to the left of the origin. Find the distance between them. 2

22. The height of water in a tank is $h(t) = 3 - 0.5t$ metres after t months. Find the height after 5 months and the time when the tank is empty. 2

23. Locate $\frac{3}{4}$ on the number line by describing the steps. 2

24. Factorise $x^2 + 11x + 30$ by splitting the middle term. 2

25. In quadrilateral ABCD, $\angle A = 100^\circ$ and $\angle C = 80^\circ$. Is it cyclic? Give a reason. 2

26. The area of a parallelogram is 96 cm^2 and its base is 12 cm. Find its height. 2

27. Why is tossing a coin considered a fair way to decide which team bats first? 2

28. Find the 10th and 26th terms of the AP $3, 8, 13, 18, \ldots$ 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Show that $A(3,\,0)$, $B(6,\,4)$ and $C(-1,\,3)$ are the vertices of an isosceles right-angled triangle. 3

30. Given $y = 2x + 1$, complete the table: $x = 1, 2, 5, 7, 9, 12, 20$ and check which of the points lies on the line: $(4, 9)$ or $(6, 14)$. 3

31. Show that $\frac{2}{\sqrt{3}}$ is irrational given that $\sqrt{3}$ is irrational. 3

32. Fill the blanks: $(px + a)(qx + b) = (__)x^2 + (__)x + __$. Verify with $(2x + 3)(3x + 1)$. 3

33. The circumferences of two circles are in the ratio 5 : 4. Find the ratio of their areas. 3

34. A day of the week is chosen at random. What is the probability that it is a weekend day? Is a random pick of 'a day starting with T' more or less likely? 3

35. The angles A and C of a cyclic quadrilateral are in the ratio 4 : 5. Find them. 3

OR

Three friends stand at three points of a park that are not in a line. Where should a fountain be placed so that it is equally far from all three? Justify. 3

36. Show that $1, 3, 6, 10, \ldots$ is neither an AP nor a GP. 3

OR

A bacterial colony of 500 doubles every hour. How many bacteria are there after 6 hours? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** Mādhava's series gives $\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$ and Āryabhaṭa gave $\pi \approx 3.1416$. 5

(i) Is π rational?

(ii) Compute $4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}\right)$ and compare with 3.1416.

(iii) Name the mathematician who proved π irrational and the year.

OR

Convert into $\frac{p}{q}$: 5

(i) 12.6

(ii) 0.0120 (terminating)

(iii) 3.052 (terminating)

(iv) $0.\overline{23}$

(v) $2.0\overline{5}$.

38. (i) Prove that an angle inscribed in a semicircle is 90°. 5

(ii) Chord AB subtends $\angle ACB = 62^\circ$ at C on the major arc. Find the angle AB subtends at the centre and at a point D on the minor arc.

OR

Case study: In a circle with centre O, AB is a diameter and C, D are points on the circle on opposite sides of AB. $\angle CAB = 30^\circ$ and $\angle ABD = 50^\circ$. 5

(i) Find $\angle ACB$ and $\angle ADB$.

(ii) Find $\angle ABC$.

(iii) Find $\angle BAD$.

(iv) State whether ACBD is cyclic and why.

39. Prove, using the inscribed and circumscribed regular hexagons of a circle of radius r, that $3 < \pi < 2\sqrt{3}$. Why is Archimedes' extension to 96 sides useful? 5

OR

Two congruent circles of radius r with centres A and B each pass through the other's centre and meet at C and D. 5

(i) Prove that $\triangle ABC$ is equilateral.

(ii) Find $\angle CAD$.

(iii) Find the perimeter of the outer boundary of the figure.

40. (i) Show that the sum of the first n odd numbers is n^2. 5

(ii) Find $1 + 3 + 5 + \cdots + 49$.

(iii) Express 144 as the sum of consecutive odd numbers starting at 1.

OR

(i) Prove that $t_n + t_{n-1} = n^2$ for triangular numbers. 5

(ii) Verify for $n = 10$.

(iii) Find the 17th and 80th triangular numbers.

END OF QUESTION PAPER **Total: 80 Marks**

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